

CEREBRO: A Subject-Conditioned Transformer for Visual EEG Encoding

Foundation-model image embeddings and cross-hardware transfer to consumer-grade EEG

Longjin Che, Ian Gloege, Bala Shukla, and Rishav Roy

1 Introduction

Our project asks whether pretrained foundation-model image embeddings can be used to predict time-resolved human EEG responses to natural images. The original proposal targeted multimodal prediction of peripheral physiological signals, especially heart-rate and electrodermal activity, from AMIGOS-style video/audio stimuli (Miranda-Correa et al. 2021), using encoders such as VideoMAE and CLAP for video and audio (Tong et al. 2022; Elizalde et al. 2023). During implementation, we pivoted to a narrower but closely related neural-encoding problem: visual EEG prediction from THINGS-EEG2 and transfer to the lower-cost Alljoined-1.6M EEG corpus. This pivot keeps the central scientific question - whether pretrained representations encode latent stimulus structure that aligns with biological responses - while making the two-month project feasible with the datasets and compute available.

The final hypothesis is:

A subject-conditioned transformer trained jointly across multiple subjects, using fixed pretrained visual embeddings, will predict image-evoked EEG more accurately than per-subject or linear baselines. If the learned encoder captures stimulus-locked neural structure rather than only dataset-specific artifacts, then a frozen THINGS-EEG2 encoder should also transfer to consumer-grade Alljoined EEG after only lightweight subject-head adaptation.

This hypothesis is motivated by TRIBE v2, which frames brain modeling as an encoding problem and uses a tri-modal foundation model to predict fMRI responses to sight, sound, and language (d’Ascoli et al. 2026). We test a smaller, EEG-specific analogue: can multi-subject pooling and a subject-specific readout help in the high-temporal-resolution, low signal-to-noise setting of event-related EEG? The main architectural addition is an RSVP context window: each target image response is predicted from the target image embedding plus the five preceding and five following image embeddings, allowing the model to condition on local temporal context in the rapid serial visual presentation paradigm.

2 Related Work

Foundation-model encoding of brain activity. TRIBE v2 shows how pretrained multimodal representations can be mapped to brain activity while using subject-specific components to account for individual variation (d’Ascoli et al. 2026). Our work adopts this logic but changes the measurement regime from slow, spatially rich fMRI to fast, noisier EEG. This makes the project a useful stress test for whether multi-subject foundation-model features help outside the original fMRI setting.

Large-scale visual EEG datasets. THINGS-EEG2 is a standard benchmark for modeling visual object recognition from EEG. It contains 10 participants, 82,160 trials per participant, and 16,740 image conditions, enabling image-level train/test splits and repeated-trial averaging (Gifford et al. 2022). Alljoined-1.6M was designed as a lower-cost companion dataset: it records visual RSVP EEG from 20 participants using a 32-channel consumer-grade EMOTIV FLEX2 system and uses the same stimulus family, which makes it unusually well suited for cross-hardware transfer experiments (Xu et al. 2025).

Pretrained visual embeddings. We compare CLIP ViT-L/14 and DINOv2-Large. CLIP learns visual features through image-text contrastive supervision and has strong zero-shot semantic transfer (Radford et al. 2021). DINOv2 learns general-purpose image features without labels and is a strong self-supervised vision baseline (Oquab et al. 2024). The comparison is scientifically useful: if CLIP wins, the relevant EEG signal may be more semantic/category-like; if DINOv2 wins, the signal may be more low-level or perceptual.

Robust multimodal physiological modeling. ADAPT studies physiological-change detection under missing modalities by aligning modalities to an anchor representation and using a masked multimodal transformer (Mordacq et al. 2024). We did not implement ADAPT’s full multimodal setup, but it influenced two design choices: treating the strongest available representation as an anchor, and using dropout/null pathways to make the model robust when subject identity or sensor format differs at transfer time. Other brainstormed directions, such as ARC-AGI-style reasoning, quantization, and world-model systems, were less directly connected to the final EEG encoding question and are therefore not treated as core related work.

3 Methods

3.1 Data and preprocessing

For THINGS-EEG2, we used preprocessed EEG epochs for 10 subjects. Each training array contains repeated responses to 16,540 training images, and each test array contains repeated responses to 200 held-out test images. We excluded the stimulus marker channel, averaged over repeated presentations, and modeled the remaining EEG as a channel-by-time matrix. For the full THINGS-EEG2 experiments this produced 63 EEG channels and 100 time samples per image. The prediction target for a subject and image is therefore $Y \in \mathbb{R}^{C \times T}$.

Image features were extracted once and kept fixed. We used CLIP ViT-L/14 embeddings with dimension 768 and DINOv2-Large CLS embeddings with dimension 1024. The attached embeddings have shapes (16540, 768) and (200, 768) for CLIP, and (16540, 1024) and (200, 1024) for DINOv2, matching the train/test image counts. We also used a concept-level 90/10 train/validation split, so no image concept appears in both training and validation.

For Alljoined transfer, we used the first 5 available Alljoined subjects. We averaged repeated trials per image, cropped the post-stimulus window, resampled to 80 post-stimulus timepoints, reordered channels to the 32-channel intersection with THINGS-EEG2, and prepended a zero baseline so the input shape matched the THINGS-EEG2 model’s time dimension. All Alljoined correlations reported below are computed only on the post-stimulus timepoints.

3.2 Channel spaces

The project uses three channel configurations, which must be distinguished carefully. First, the main THINGS-EEG2 model uses the 63 EEG channels available after removing the marker channel. Second, an exploratory 28-channel visual/posterior-heavy mask improves in-domain THINGS-EEG2 performance by focusing capacity on visually responsive sensors. Third, cross-hardware transfer uses the true 32-channel THINGS-EEG2/Alljoined intersection: Cz, FCz, AFz, Fp1/2, F1/2/5/6, CP1/2/3/4/5/6, P1/2/3/4/5/6/7/8/Pz, PO3/4/7/8/POz, and O1/O2/Oz. Only the 32-channel intersection model is used for the Alljoined transfer results.

3.3 Models

All transformer models use a frozen image embedding followed by a learned projection to $d = 384$, learned positional embeddings over an 11-token context, and four pre-norm transformer encoder layers with six attention heads (Vaswani et al. 2017). The context contains the target image token plus five previous and five following image tokens, with zero-padding at sequence boundaries.

The multi-subject model uses a shared transformer backbone and a subject-indexed linear readout. Let $h \in \mathbb{R}^{384}$ be the target-token transformer output. The prediction for subject s is

$$\hat{Y}_s = \text{reshape}(W_s h + b_s, C, T),$$

where $W \in \mathbb{R}^{(N+1) \times CT \times 384}$ and $b \in \mathbb{R}^{(N+1) \times CT}$. The final block, index N under zero-based indexing, is a null-subject pathway sampled with probability 0.1 during training. This subject dropout is intended to make the backbone usable for unseen subjects. We compare against two baselines: a per-subject transformer with the same general form but no cross-subject pooling, and a per-subject linear regression head from image embedding directly to flattened EEG.

3.4 Training and evaluation

Models were trained with AdamW, learning rate 10^{-4} for the transformer experiments, weight decay 0.01, cosine scheduling with warmup, batch size 64, and early stopping on validation Pearson correlation. The main CLIP multi-subject result averages three independent random seeds: multi_subject_clip_seed1337, multi_subject_clip_seed2026, multi_subject_clip_seed42. The main metric is Pearson correlation between predicted and observed EEG, computed per channel-time cell over the held-out test images and then averaged over cells and subjects. For Alljoined, we report both all-cell correlation and posterior-channel correlation; posterior correlation is the transfer headline because visual responses are strongest over parietal/occipital sensors.

For cross-hardware transfer, we initialized from the 32-channel THINGS-EEG2 intersection model and compared four Alljoined protocols: scratch ridge regression from CLIP embeddings, Variant A zero-shot prediction through the null-subject pathway, Variant B frozen-encoder subject-block fine-tuning, and Variant C full fine-tuning for one epoch. Variant B trains only a fresh subject readout block initialized from the mean THINGS-EEG2 subject block.

4 Results

4.1 Multi-subject pooling improves THINGS-EEG2 prediction

Table 1 summarizes the main THINGS-EEG2 results. The 63-channel multi-subject CLIP transformer is the main comparison to the per-subject and linear baselines. It reaches mean $R = 0.0912$, compared with $R = 0.0416$ for the linear CLIP baseline and $R = 0.0394$ for the tuned per-subject transformer. Paired tests across the 10 subjects give $p = 1.25 \times 10^{-8}$ for multi-subject CLIP versus linear and $p = 3.71 \times 10^{-8}$ for multi-subject CLIP versus per-subject transformer. Linear and per-subject transformer performance are not distinguishable ($p = 0.549$), consistent with the idea that a per-subject transformer is too data-hungry for roughly 14k training examples per subject.

Table 1: THINGS-EEG2 test Pearson correlation. Correlations are averaged over channel-time cells and subjects. The 28-channel mask is an in-domain visual-channel model; the 32-channel row is the model used for cross-hardware transfer.

Model	Embedding / channels	Mean R	SD across subjects
Linear, per subject	CLIP, 63 ch	0.0416	0.0083
Transformer, per subject	CLIP, 63 ch	0.0394	0.0183
Transformer, multi-subject	CLIP, 63 ch, 3-seed avg	0.0912	0.0160
Transformer, multi-subject	DINOv2, 63 ch	0.0681	0.0133
Transformer, multi-subject	CLIP, 28 ch visual mask	0.1343	0.0185
Transformer, multi-subject	CLIP, 32 ch	0.1219	0.0193
	TS2-Alljoined intersection		

The CLIP/DINOv2 comparison supports the semantic-feature hypothesis. CLIP beats DINOv2 by 0.0230 mean R across subjects ($p = 1.11 \times 10^{-8}$), suggesting that language-aligned semantic structure is more predictive of the averaged visual EEG response than DINOv2’s purely self-supervised visual representation.

The channel-restricted models also behave as expected. The 28-channel posterior-heavy mask reaches $R = 0.1343$, and the true 32-channel TS2-Alljoined intersection reaches $R = 0.1219$. This gain over the full 63-channel model is

not evidence that the larger model is worse in all respects; rather, the all-cell average is diluted by frontal/noisier sensors, while visual EEG is concentrated over posterior channels.

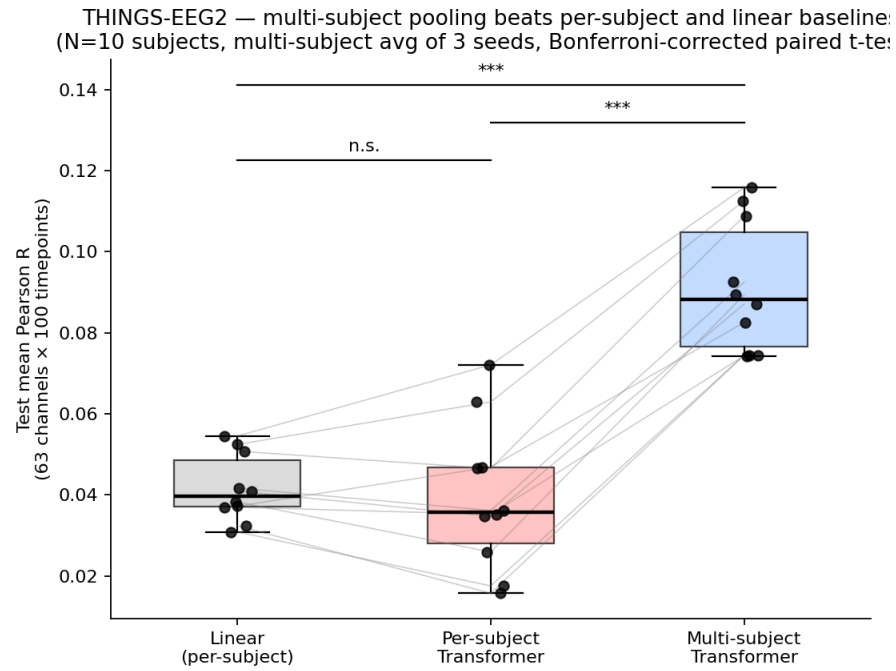


Figure 1: Multi-subject CLIP outperforms linear and per-subject baselines on THINGS-EEG2.

Multi-subject CLIP (seed 42) — Pearson R per (channel, timepoint), averaged across N=10 subjects
Global mean R = 0.089, max = 0.660

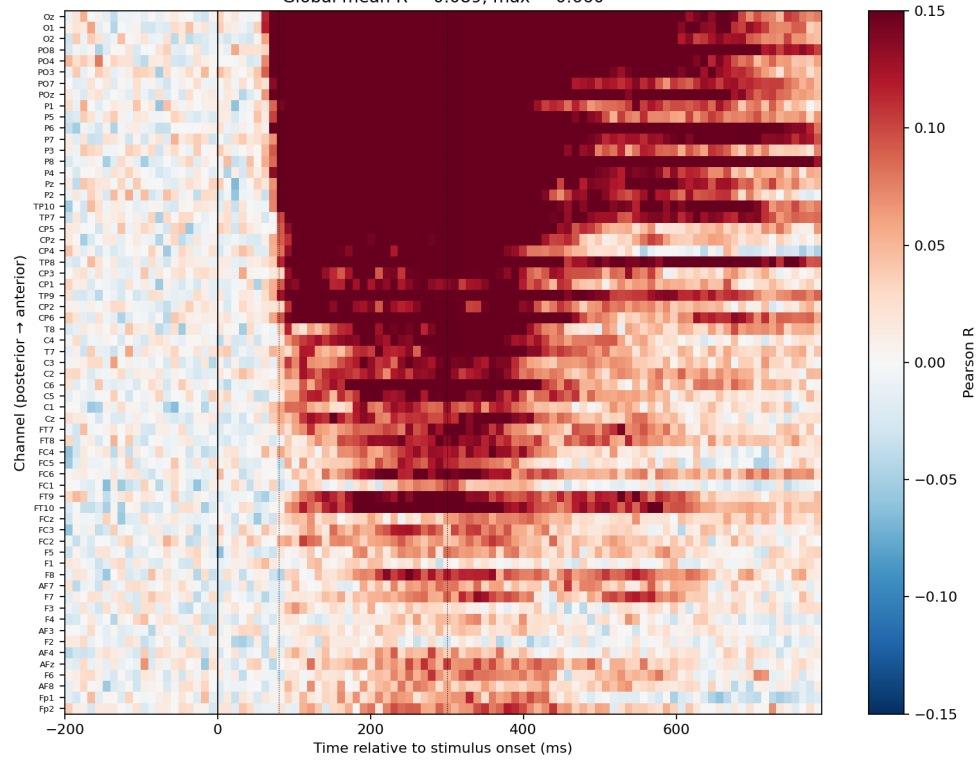


Figure 2: Channel-by-time correlation heatmap for the multi-subject CLIP model. The plotted color scale is clipped to emphasize spatial-temporal structure; the highest individual cells exceed the top of the color bar.

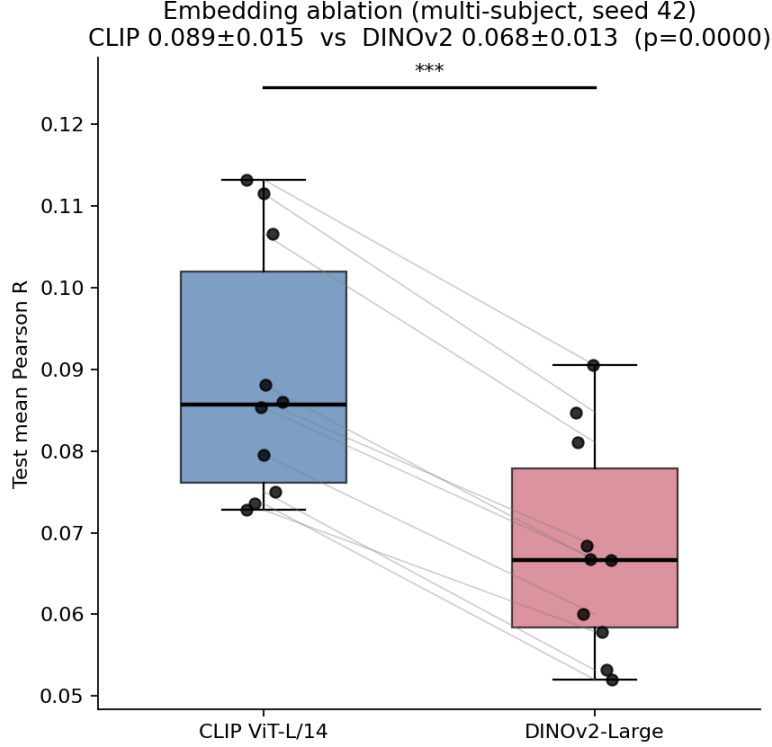


Figure 3: Embedding ablation: CLIP outperforms DINOv2 for this visual EEG encoding task.

4.2 Frozen-encoder subject-head adaptation transfers to Alljoined

Table 2 shows preliminary transfer results on 5 Alljoined subjects. The scratch ridge baseline gives posterior $R = 0.0073$. The null-subject zero-shot path improves the mean to $R = 0.0223$ but is not significant with $N = 5$ ($p = 0.301$). The strongest result is Variant B: freezing the THINGS-EEG2 encoder and training only a new Alljoined subject block gives posterior $R = 0.0299$, with an uncorrected paired t-test against ridge of $t = 3.18$, $p = 0.034$. Full fine-tuning reaches $R = 0.0235$ but is high-variance and not significant ($p = 0.336$), suggesting that full adaptation needs validation-based early stopping or a lower learning rate.

Table 2: Alljoined-1.6M posterior-channel transfer results for subjects 01-05. These are preliminary because only 5 of 20 subjects were processed.

Variant	Posterior R	SD	Uncorrected paired test vs ridge
Scratch ridge	0.0073	0.0090	–
Variant A: zero-shot null pathway	0.0223	0.0237	$t = 1.19$, $p = 0.301$
Variant B: frozen encoder + subject block	0.0299	0.0151	$t = 3.18$, $p = 0.034$
Variant C: full fine-tuning	0.0235	0.0279	$t = 1.09$, $p = 0.336$

The most conservative interpretation is that the transfer result is promising but not yet definitive. Variant B is about four times the ridge baseline in mean posterior correlation, but the sample size is small, and several transfer variants were explored. Therefore, the result should be treated as evidence that the learned encoder may transfer across hardware, not as a final claim about consumer-grade BCI reliability.

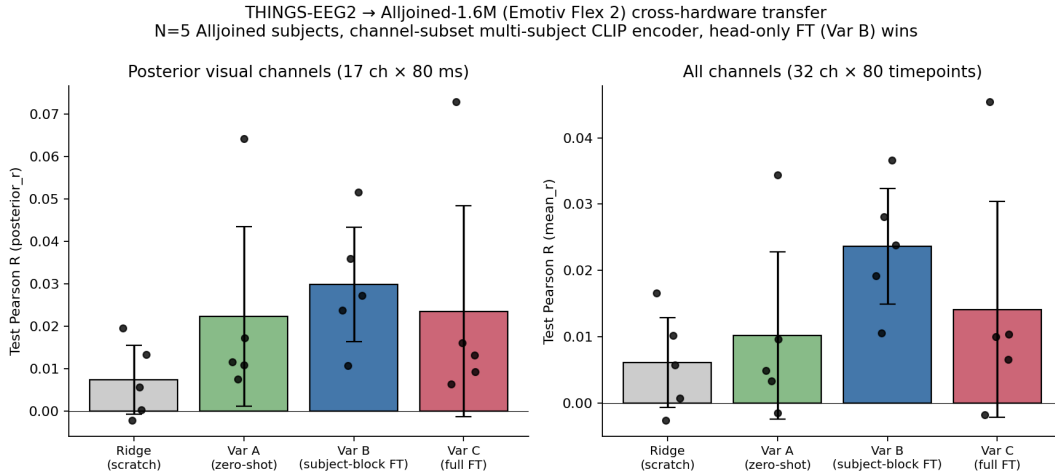


Figure 4: Cross-hardware transfer to Alljoined. Variant B is the strongest protocol, but the small subject count makes the result preliminary.

5 Discussion, Conclusions, and Future Work

The results support the main hypothesis on THINGS-EEG2: a subject-conditioned multi-subject transformer substantially outperforms both a linear baseline and a per-subject transformer. This is the clearest success of the project and directly addresses the course goal of extending ideas from foundation-model neural encoding into a new setting. The results also partially support the transfer hypothesis: frozen-encoder subject-head fine-tuning improves over scratch ridge on the first few Alljoined subjects, but this conclusion is limited by sample size and should be replicated on all 20 subjects.

The project also clarified what did not work. The per-subject transformer did not improve over linear regression, which suggests that architectural capacity alone is not helpful without enough subject-specific data. Full fine-tuning on Alljoined was unstable, probably because updating the entire encoder on one subject can overfit quickly. Finally, the original ECG/EDA multimodal goal was not completed; the final contribution is instead a focused EEG encoding and transfer study. This should be presented as a deliberate scope change, not as completed AMIGOS peripheral-physiology work.

Several future directions follow naturally. First, the Alljoined experiment should be extended from five to all twenty subjects, with a pre-registered choice of the posterior-channel metric. Second, full fine-tuning should use a validation split, early stopping, and lower-rank adaptation rather than one unconditional epoch. Third, the model should be compared with stronger spatiotemporal baselines, such as regularized temporal response functions or MNE-style encoding models. Fourth, the original multimodal motivation could be revisited by adding audio/video features and physiological targets from AMIGOS, using ADAPT-like missing-modality training for robustness (Mordacq et al. 2024).

In conclusion, the current work studies how pretrained representations align with human biological responses, develops a subject-conditioned transformer architecture for EEG, evaluates the hypothesis against baselines, and provides preliminary evidence for cross-hardware transfer to a cheaper EEG device.

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